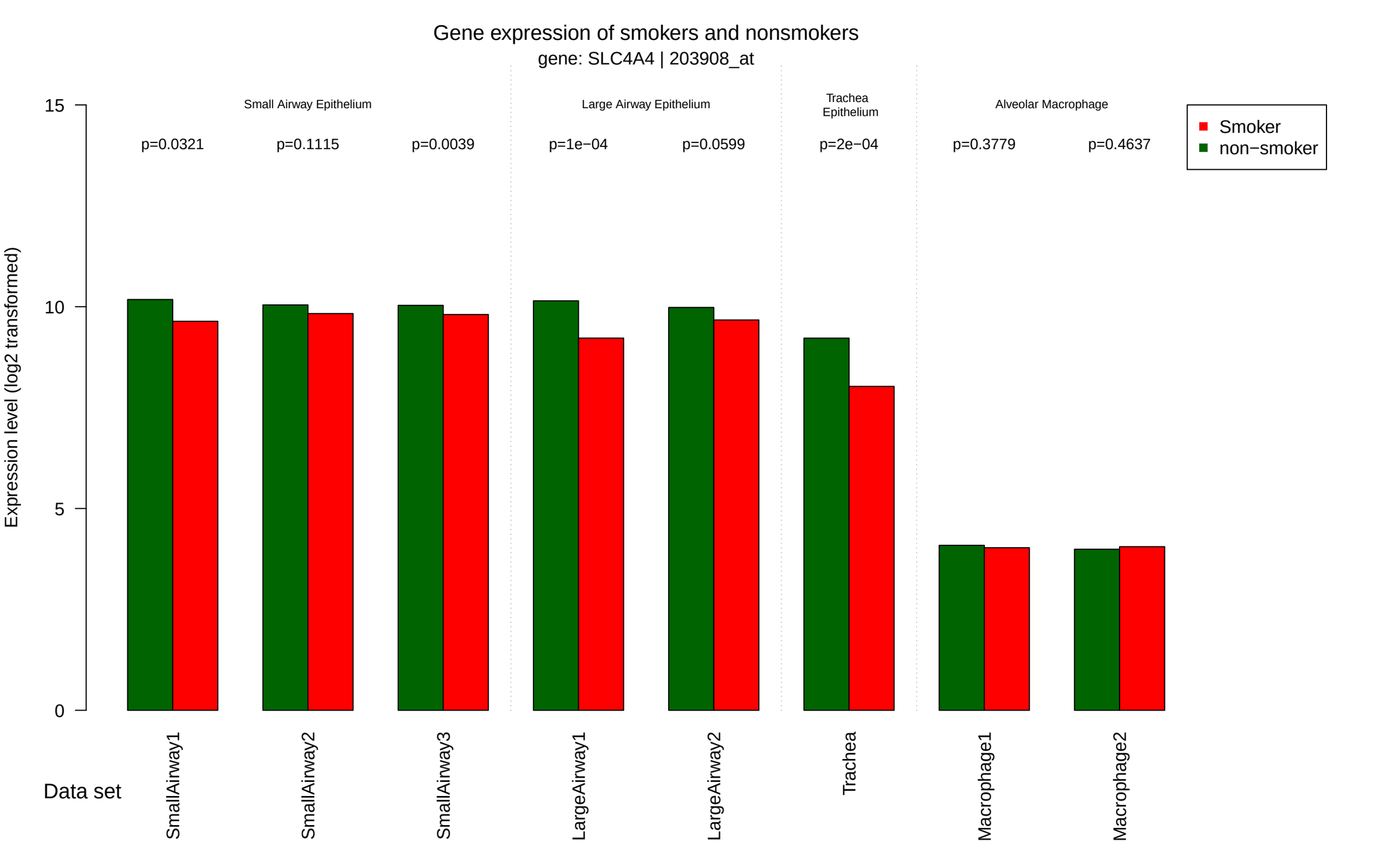
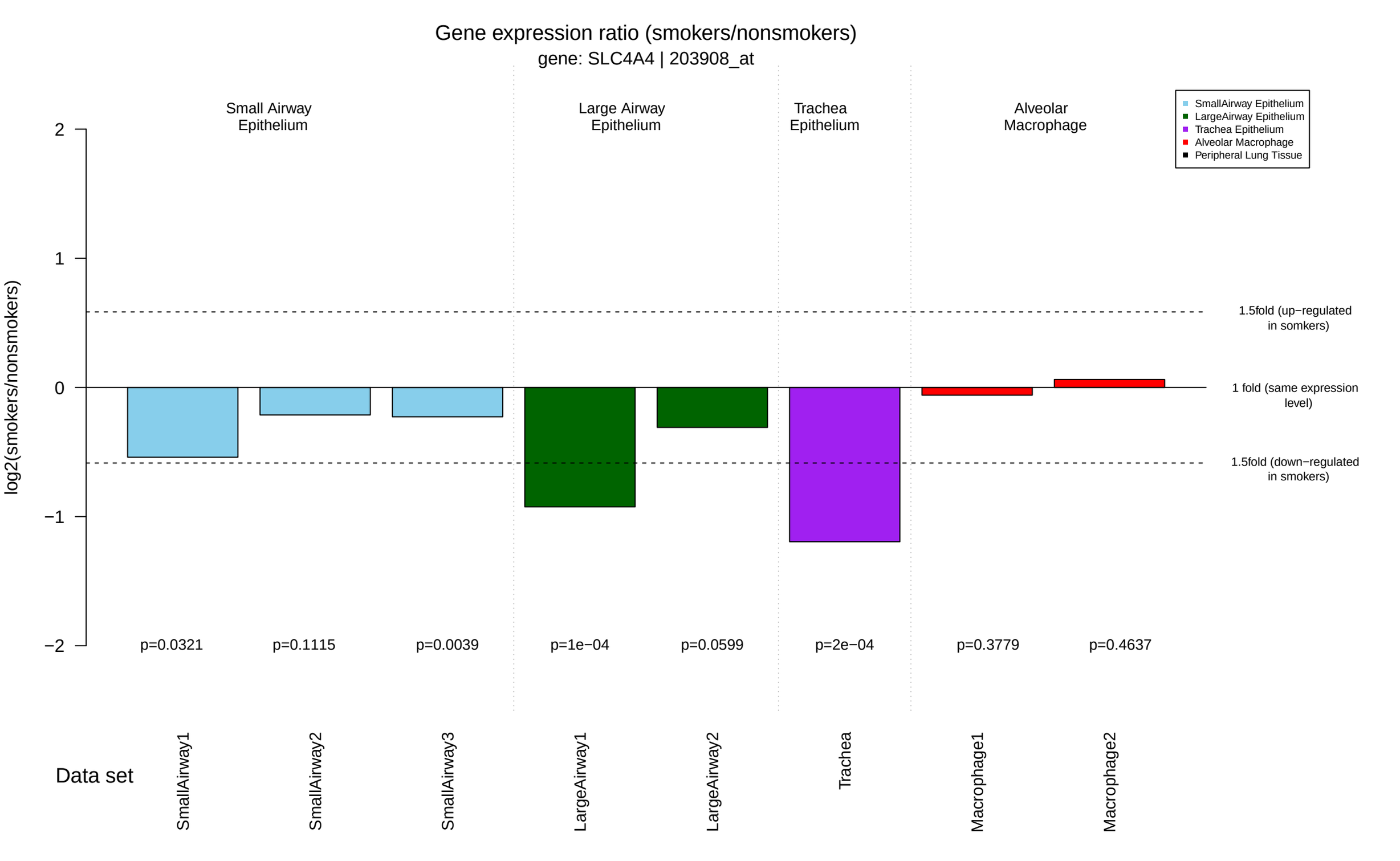
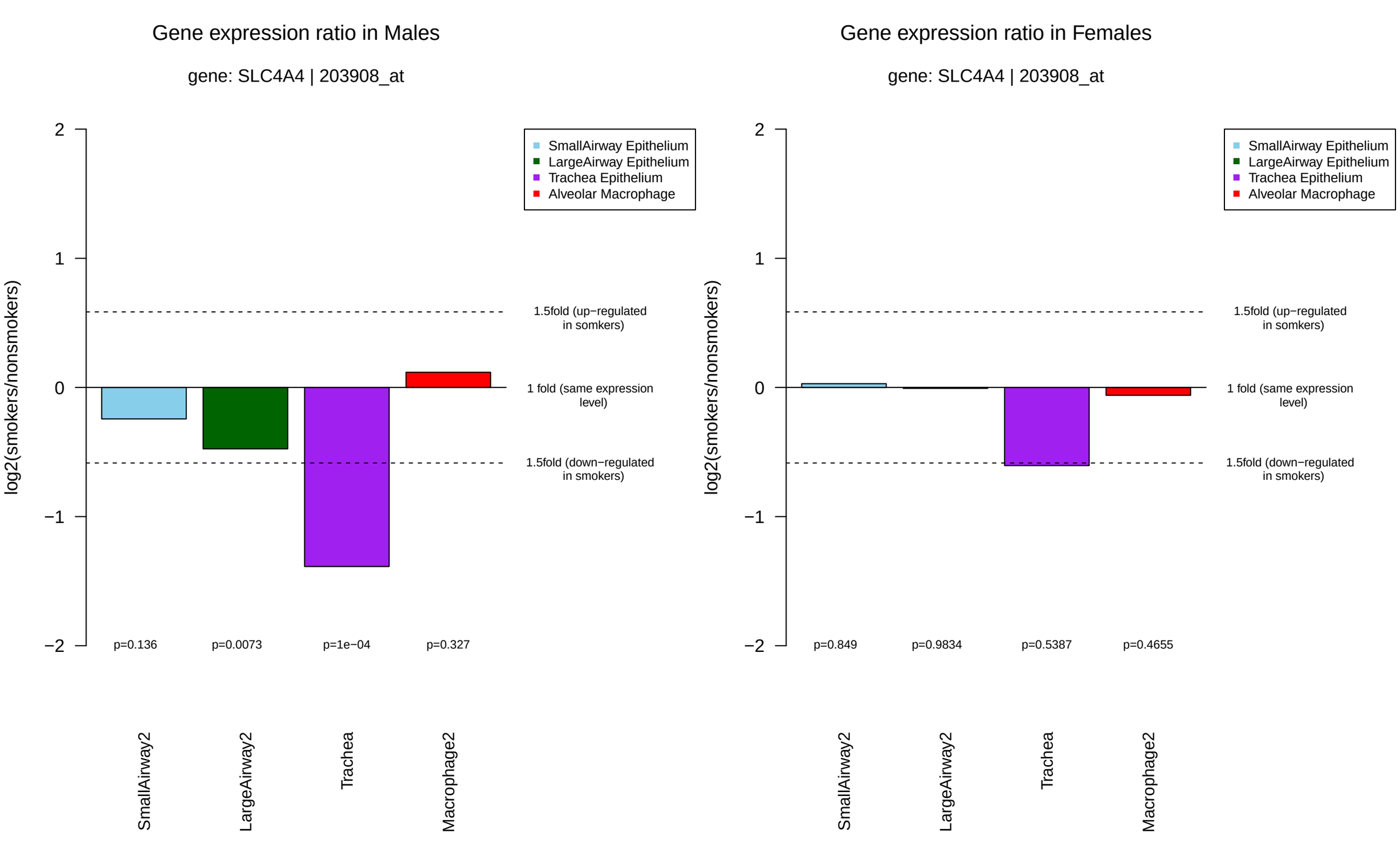
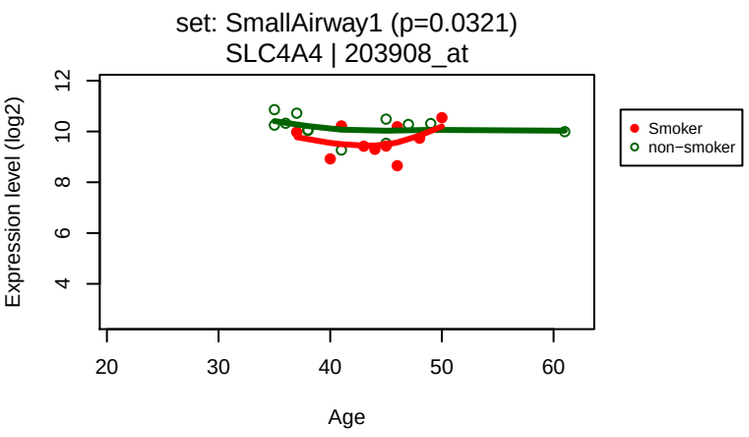
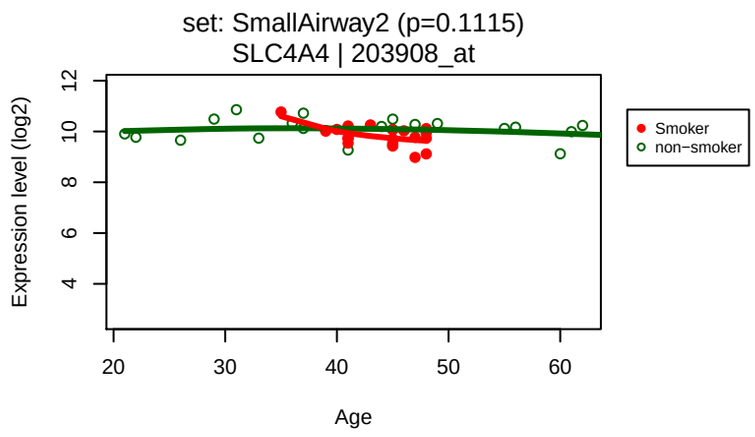
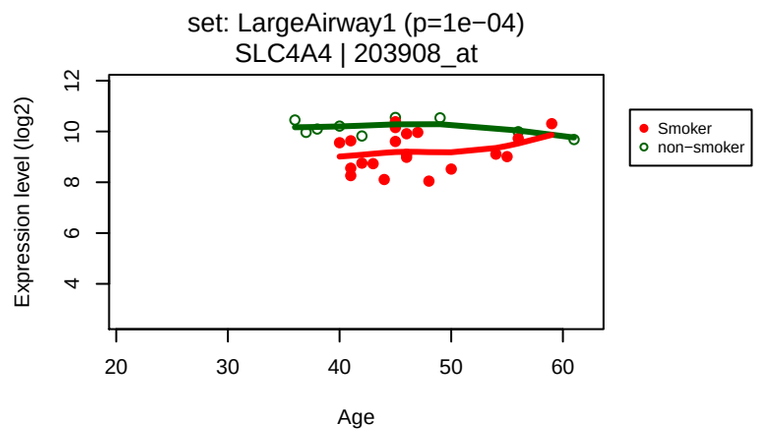

Gene expression of smokers and nonsmokers
gene: SLC4A4 | 203908_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0321
p=0.1115
p=0.0039
p=1e−04
p=0.0599
p=2e−04
p=0.3779
p=0.4637
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

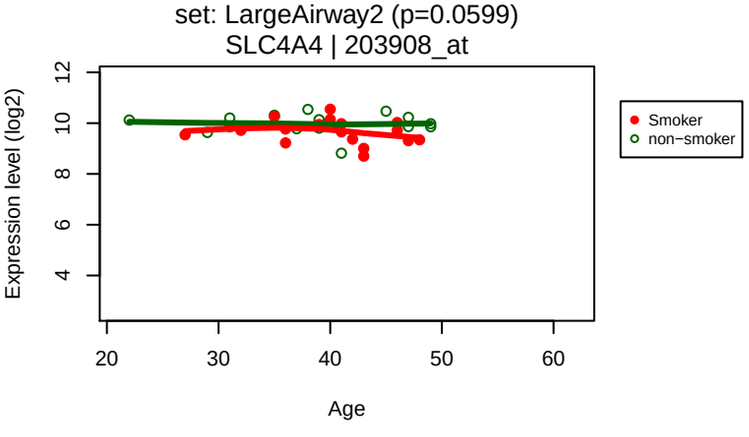

Gene expression ratio (smokers/nonsmokers)
gene: SLC4A4 | 203908_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0321
p=0.1115
p=0.0039
p=1e−04
p=0.0599
p=2e−04
p=0.3779
p=0.4637
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

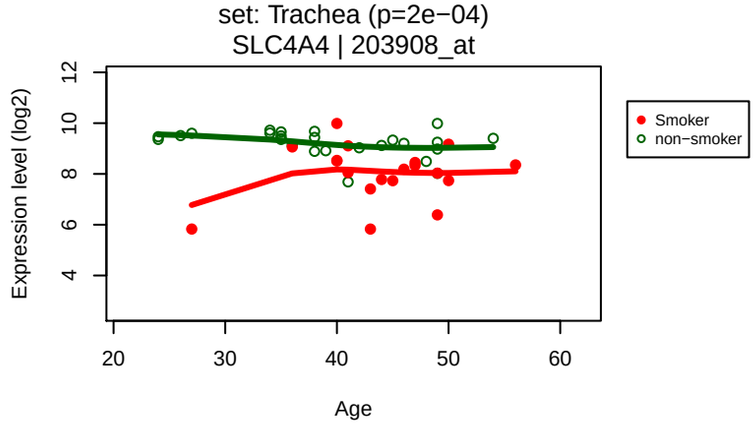

Gene expression ratio in Males
gene: SLC4A4 | 203908_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.136
p=0.0073
p=1e−04
p=0.327
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SLC4A4 | 203908_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.849
p=0.9834
p=0.5387
p=0.4655
SmallAirway2
LargeAirway2
Trachea
Macrophage2

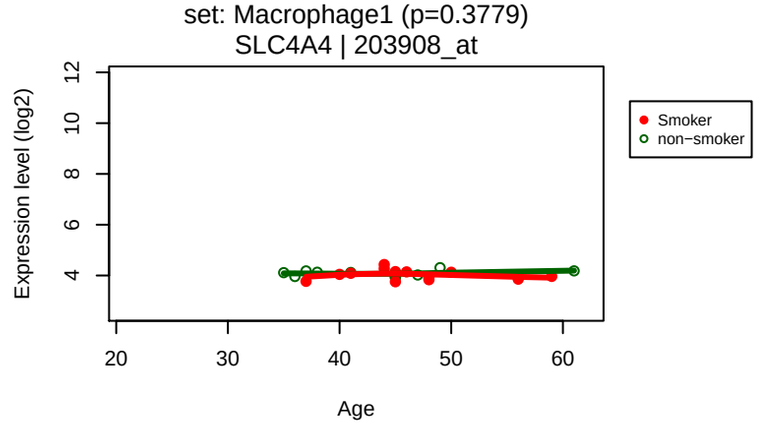

set: SmallAirway1 (p=0.0321)
SLC4A4 | 203908_at
Expression level (log2)
Age
Smoker
non−smoker

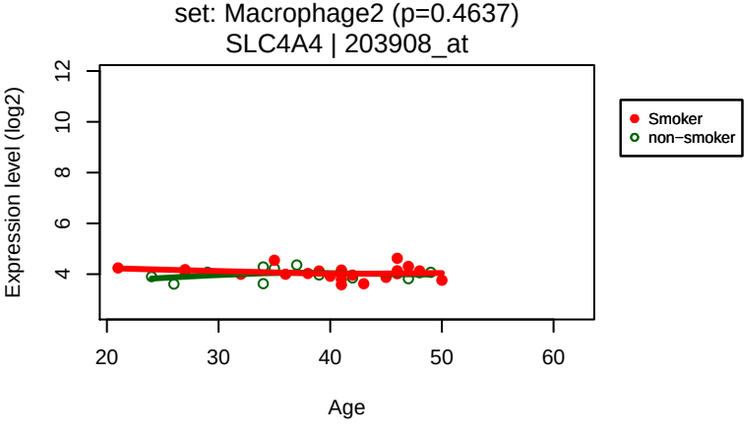

set: SmallAirway2 (p=0.1115)
SLC4A4 | 203908_at
Expression level (log2)
Age
Smoker
non−smoker

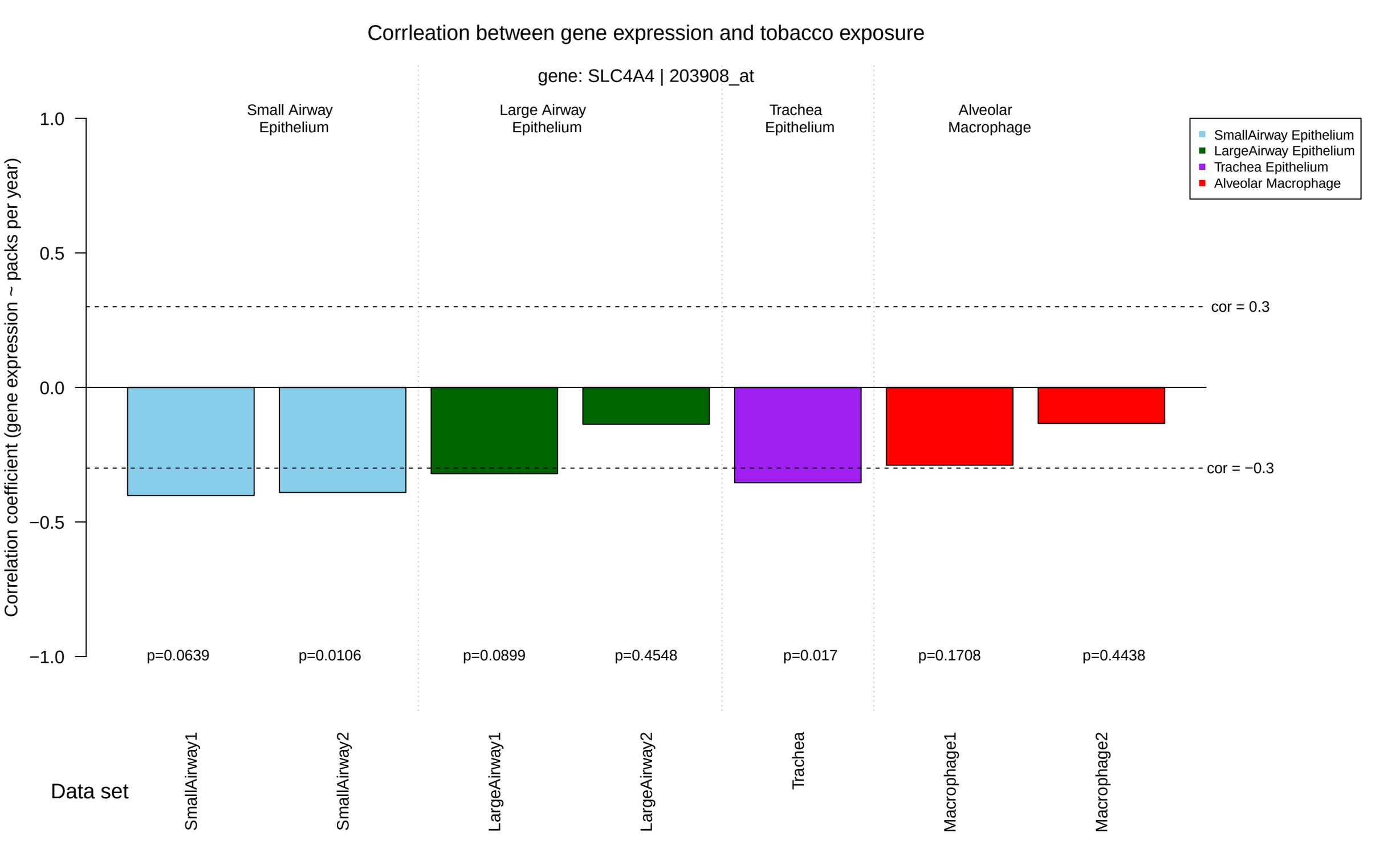

set: LargeAirway1 (p=1e−04)
SLC4A4 | 203908_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0599)
SLC4A4 | 203908_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=2e−04)
SLC4A4 | 203908_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3779)
SLC4A4 | 203908_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4637)
SLC4A4 | 203908_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SLC4A4 | 203908_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0639
p=0.0106
p=0.0899
p=0.4548
p=0.017
p=0.1708
p=0.4438
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2